

SSH Brute Force Attack

Attack Type	SSH Brute Force (T1110)
Target	KALI-UTM — 10.11.3.106
Attacker Host	Mac M4 Host — 10.11.3.152
Date	April 21, 2026
Wazuh Rule	5712 — Level 10 (Medium-High)
MITRE Tactic	Credential Access (TA0006)
MITRE Technique	T1110 — Brute Force / T1110.001
Result	DETECTED — Alert fired after 8+ attempts in 120s
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Course	DAE Cybersecurity Cohort — Stafford, CT

1. Executive Summary

This report documents the full incident timeline of an SSH Brute Force attack simulated against the Kali Linux endpoint (KALI-UTM) as part of Phase 2 of The Sentinel Hybrid-SOC project. The attack was deliberately triggered to validate that the Wazuh SIEM platform correctly detects repeated failed SSH login attempts and fires a real-time alert.

The simulation confirmed that Wazuh Rule 5712 (Level 10) fires after 8 or more failed login attempts within a 120-second window from the same source IP — exactly matching real-world brute force behavior. The full attack chain from execution to alert is documented below with timestamps and evidence.

2. Lab Environment

Wazuh Manager	Docker single-node deployment on Mac M4 Host (10.11.3.152) — v4.14.4
Wazuh Agent	KALI-UTM — Agent ID 001 — Kali Linux 2023.3 ARM64 on UTM
Target IP	10.11.3.106 — Kali Linux (SSH listener on port 22)
Attacker IP	10.11.3.152 — Mac M4 Host (ran brute force loop)
Network	10.11.3.0/22 — Host-only subnet (UTM virtual network)
Attack Tool	Native OpenSSH client — shell loop (no special tools required)
Attack Date	April 21, 2026
Wazuh Dashboard	https://localhost (Docker-hosted, port 443)

3. Attack Overview

An SSH Brute Force attack is one of the most common techniques attackers use to gain unauthorized access to systems. Automated tools — and in this case a simple shell loop — attempt to log in using a large number of username and password combinations in rapid succession, hoping to guess valid credentials.

In this simulation, the Mac host attempted to SSH into the Kali VM 10 times using the username 'yessy' — which does not exist on the Kali system. Each attempt was denied, and the pattern of repeated failures from the same source IP triggered Wazuh's brute force detection rule.

Attack Command Used:

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for i in {1..10}; do ssh yessy@10.11.3.106; done
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This loop runs 10 SSH login attempts in succession. Each attempt uses the non-existent username 'yessy' and intentionally wrong passwords. The rapid repeated failures from the same IP address are the signature Wazuh looks for.

MITRE ATT&CK Mapping:

Field	Details
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Tactic	Credential Access (TA0006)
Technique	T1110 — Brute Force
Sub-technique	T1110.001 — Password Guessing
Platform	Linux (Kali GNU/Linux 2023.3 ARM64)
Protocol	SSH — Port 22
Adversary Goal	Gain unauthorized shell access by guessing credentials

4. Wazuh Detection Rule — Rule 5712

Before running the attack, Wazuh Rule 5712 was reviewed to understand exactly what threshold would trigger the alert. This is standard SOC analyst practice: know your detection rules so you can verify they are working correctly.

Rule ID	5712
Rule Level	10 — Medium-High severity
Description	sshd: brute force trying to get access to the system. Non-existent user.
Rule File	0095-sshd_rules.xml
Detection Groups	authentication_failures, syslog, sshd
Trigger Frequency	8 failed attempts within 120 seconds from same source IP
Same Source IP	true — all attempts must come from the same attacker IP
Parent Rule	5710 — sshd: Attempt to login using a non-existent user
MITRE Technique	T1110 — Brute Force
MITRE Tactic	Credential Access (TA0006)
Compliance	PCI DSS 11.4, 10.2.4, 10.2.5 HIPAA 164.312.b NIST-800-53 SI.4, AU.14, AC.7

How Rule 5712 Works: Rule 5710 fires on every single failed SSH attempt using a non-existent username. Rule 5712 is a frequency rule that builds on 5710 — once 8 instances of Rule 5710 fire from the same source IP within 120 seconds, Rule 5712 escalates the alert to Level 10 and classifies it as a brute force attack.

5. Incident Timeline

The following timeline documents each stage of the attack from setup through detection, with the exact sequence of events as they occurred on April 21, 2026.

Time (PDT)	Phase	Event	System	Evidence
12:10:54	Setup	SSH service started on Kali Linux	KALI-UTM	systemctl status ssh — active (running)
12:10:54	Setup	Kali IP confirmed: 10.11.3.106 on port 22	KALI-UTM	ifconfig — eth0 inet 10.11.3.106
12:11:xx	Pre-Attack	Rule 5712 reviewed in Wazuh Rules panel	Wazuh	Rules panel — Level 10, frequency 8/120s
12:15:xx	Attack	Brute force loop started from Mac host	Mac Host	for i in {1..10}; do ssh yessy@10.11.3.106; done
12:15:xx	Attack	Attempt 1 — Permission denied (non-existent user)	SSH / Kali	yessy@10.11.3.106: Permission denied
12:15:xx	Attack	Attempts 2–7 — Repeated Permission denied responses	SSH / Kali	Repeated: Permission denied (publickey,password)
12:15:xx	Attack	Attempt 8 threshold crossed — Rule 5712 triggered	Wazuh	Rule 5712 Level 10 alert fires in Wazuh dashboard
12:15:xx	Attack	Attempts 9–10 continue — connection closed	SSH / Kali	Connection closed by 10.11.3.106 port 22
~12:15	Detection	Wazuh alert visible in Security Events dashboard	Wazuh	29 medium + 47 low alerts — Rule 5712 in event list
~12:15	Detection	Threat Hunting module confirms full attack timeline	Wazuh	authentication_failures group — 26 events logged
~12:15	Analysis	Wireshark capture confirms brute force pattern	Wireshark	SYN→SYN-ACK→ACK→AUTH-FAIL→RST repeated 28 times

6. Detection Evidence

The following evidence was captured from the Wazuh dashboard and terminal during and after the attack. Each piece of evidence is referenced by the screenshot taken during the lab exercise.

Evidence Item	Source	What It Proves
Wazuh Dashboard — 1 active agent (KALI-UTM) 29 medium + 47 low alerts	Wazuh Dashboard Figure 1	Agent is actively monitored. Multiple security events generated by the attack.
SSH service active and running on port 22 Kali IP: 10.11.3.106	Kali Terminal Figure 2	Attack surface confirmed — Kali was a valid SSH target.
Rule 5712 detail in Wazuh Rules panel Level 10, freq=8/120s, MITRE T1110	Wazuh Rules Figure 3	Detection rule verified before attack — threshold understood.
Raw XML rule file 0095-sshd_rules.xml Rules 5700–5712 visible	Wazuh Manager Figure 4	Rule chain confirmed — 5710 (single fail) → 5712 (brute force aggregate).
SSH loop output: Permission denied (publickey,password) Connection closed by 10.11.3.106 port 22	Mac Terminal Figure 5	Attack executed successfully. 10 failed attempts confirmed from output.

Wazuh Security Events — Rule 5712 alert visible Level 10 event in alert list	Wazuh Dashboard Figure 6	Rule 5712 fired. Alert logged with full context: timestamp, rule, agent.
Threat Hunting — authentication_failures group 26 events total logged	Wazuh Threat Hunting	Full attack timeline traceable in Wazuh. Every failed attempt logged.
Wireshark capture: SYN→SYN-ACK→ACK→AUTH-FAIL→RST <100ms between attempts	Wireshark Figure 7	Network forensics confirm automated brute force pattern at packet level.

7. Containment & Response Actions

This section documents the response actions that were identified as a result of the SSH brute force detection. In a production SOC environment, these steps would be executed immediately upon alert. In this lab, they are documented as the recommended response procedure.

Step	Action	Tool / Method	Priority
1	Identify source IP of brute force attempts	Wazuh Security Events – filter by Rule 5712	Immediate
2	Block source IP at firewall / host level	iptables -A INPUT -s -j DROP	Immediate
3	Review /var/log/auth.log for additional attempts	SSH into Kali: sudo cat /var/log/auth.log grep Failed	Within 5 min
4	Check for successful logins from same IP	Wazuh Threat Hunting – filter by source IP, Rule 5715	Within 5 min
5	Disable password-based SSH authentication	Edit /etc/ssh/sshd_config: PasswordAuthentication no	Within 1 hour
6	Enable SSH key-only authentication	Generate key pair, add to authorized_keys	Within 1 hour
7	Enable Wazuh Active Response for auto-blocking	Configure active-response block in ossec.conf	Within 24 hours
8	Document incident in incident log	Complete Incident Response Summary template	Within 24 hours

8. Lessons Learned

1	SSH brute force is a real, common threat	Automated bots constantly scan the internet for open port 22. This simulation used a simple shell loop — no special tools. Any exposed SSH port faces this risk continuously.
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2	SIEM tools like Wazuh are essential	Without Wazuh, these failed login attempts would be invisible unless someone manually reviewed <code>/var/log/auth.log</code> . Wazuh provides real-time visibility with zero manual effort.
3	Rule tuning avoids alert fatigue	The 120-second window and <code>same_source_ip</code> requirement in Rule 5712 prevent false positives from normal users mistyping their password once. Smart thresholds are as important as the detection itself.
4	Compliance coverage is built-in	Detecting brute force activity automatically satisfies requirements under PCI DSS 11.4, HIPAA 164.312.b, and NIST SP 800-53 AC.7. The SIEM logs become audit evidence.
5	Defense in depth matters	Even though the attack was detected, the recommended hardening steps (key-only auth, fail2ban, Wazuh Active Response) would prevent the attack from even reaching the detection threshold in a production environment.
6	Wireshark provides forensic depth	The packet capture confirmed the brute force pattern at the network level — SYN handshakes repeating under 100ms apart. This is evidence that a tool review alone (Wazuh logs) combined with packet analysis gives a complete forensic picture.

9. Outcome & SOC Value

Detection Result	SUCCESS — Wazuh Rule 5712 fired correctly
Alert Level	Level 10 — Medium-High severity
Time to Detection	Immediate — alert fires in real time as threshold is crossed
Events Logged	26 authentication_failure events captured in Wazuh Threat Hunting
Forensic Evidence	Wazuh alerts + <code>/var/log/auth.log</code> + Wireshark packet capture
MITRE Coverage	TA0006 Credential Access — T1110 Brute Force — T1110.001 Password Guessing
Compliance Coverage	PCI DSS 11.4 HIPAA 164.312.b NIST-800-53 SI.4, AU.14, AC.7
Analyst	Yessy Vasquez — DAE Cybersecurity Cohort — May 2026

This incident timeline demonstrates that The Sentinel Hybrid-SOC successfully detected a real-world class attack technique using open-source tooling on consumer hardware. The Wazuh SIEM platform provided immediate, actionable visibility into the attack with full MITRE ATT&CK context and compliance evidence — exactly what a Security Operations Center is designed to deliver.